AI Builder Cup 2026 · Theme 5: Future of Work & Enterprise Productivity

FlowHub

One hub for everything your company sells, with Gemini agents that turn it into tailored offers.

FlowHub is the hub for everything a company sells, with an agent team built in. Staff keep everything the company sells in one place (offerings, pricing, proof, constraints and partners), extracted from the documents they already have and linked back to the source. When an opportunity arrives, Gemini agents research the prospect with cited sources, map its gaps to the company's own offerings, price the fit from the rate cards and draft the proposal into Google Docs. A person approves each stage.

5 steps
one workflow every business repeats by hand

1 hub
for the whole team: Owners, Editors, Viewers

6 agents
built on Google's Agent Development Kit

100% cited
every research claim links to its source

Team: [Team name] · [Member 1, role] · [Member 2, role]
Problem statement category: Future of Work & Enterprise Productivity
Date: September 2026
Deployed prototype: [Cloud Run URL] · **Repository:** [public GitHub URL] · **Video:** [YouTube URL]

1. The problem

Every business has the same recurring moment: an opportunity appears and someone has to show up prepared with a tailored offer. A discovery call lands in the calendar. A lead emails. An investor agrees to a coffee. A tender is published. The work behind each is identical and it is done from scratch every time.

1. **Know yourself.** What do we actually offer, at what price, with what proof? The answer lives in scattered slide decks, rate cards and old proposals.
2. **Know them.** Who is this prospect, what are they doing, where are they stuck? Usually twenty minutes of browser tabs, or nothing at all.
3. **Find the fit.** Which of our things solves which of their gaps, and what would we customise? This is the step that wins or loses the deal, and it happens in someone's head.
4. **Make the ask.** A brief for the meeting, then a proposal, quote, deck or reply. Hours of writing that starts from last quarter's document.
5. **Close the loop.** Follow up, and learn from whether it was won or lost. Almost nobody does this systematically.

Who feels it most

Large enterprises have sales engineers, bid teams and a CRM full of AI features. Most companies in the JAPAC region do not. In SMEs, startups and specialist integrators, what the company sells lives in scattered decks, spreadsheets and people's heads, and selling is done by the same people who deliver the work. Preparation competes with billable hours, knowledge leaves with the people who leave, and offers go out generic.

> **The gap in today's tools.** Each step has its own product category: meeting-prep tools that produce a brief but know nothing about what you sell; RFP suites that know your content library but only speak tender; proposal writers that format a document from a template; roleplay tools built for training sales development reps. All of them assume you already live in a CRM. Nobody gives a company one grounded source of truth for what it sells, and then uses it to drive all five steps.

Why this matters for the theme

Theme requirement	How FlowHub addresses it
Access knowledge	A structured, searchable model of the company's own offering, plus grounded research on the prospect with citations.
Collaborate	One hub for the whole team, with Owner, Editor and Viewer roles, and shared deal rooms where people approve each agent step.
Make decisions	A fit map and a go/no-go recommendation with reasoning, before any writing starts.
Complete everyday work	The brief, proposal, quote and follow-up are produced as real Google Docs and Slides files, not chat text.

2. The solution

FlowHub is the hub for everything a company sells, with an agent team built in. The **hub** is where staff find everything the company sells. It is an **offering graph** of offerings, tiers, rate cards, case studies, constraints and partners, extracted from the documents the company already has, with every item linked to its source page. The **deal room** is each opportunity's page in the hub, where a team of agents works from the hub and a person approves every stage.

The user's journey

1. **Set up the hub.** Upload the deck, the rate card and a few past proposals. Gemini reads them, including scanned PDFs, and proposes the catalogue. The team confirms it in a few minutes and invites colleagues as Editors or Viewers.
2. **Open an opportunity.** Paste a calendar invite, an email, a company URL or a tender PDF. The Intake agent classifies it and extracts who, what and when.
3. **Read the brief.** The Researcher agent produces a one-page brief on the prospect with every claim linked to its source: what they do, recent news, hiring signals, technology signals, and the sector's typical pressures.
4. **Review the fit map.** The Fit Analyst matches each identified gap to an item in the offering graph, proposes customisations, flags what cannot be met, and gives a go/no-go score with reasoning. The Pricing agent adds an indicative estimate from the rate cards.
5. **Approve the artefact.** The Writer produces the persona-appropriate output as a Google Doc or Slides file. The Reviewer agent checks it for unsupported claims before the human sees it.
6. **Ask or rehearse.** Any member can ask the hub a question and get an answer cited to the catalogue. Before the meeting, the seller can rehearse out loud with Gemini Live playing the prospect.
7. **Record the outcome.** Won, lost, or no decision, with a reason. FlowHub adjusts what it emphasises for that sector next time.

One engine, three personas

Persona	Demo seller	Intake	Output the Writer produces
Startup	Stoklane, AI demand forecasting SaaS, Singapore	Customer meeting invite	Meeting brief and pilot proposal
SME	Kelapa Cloud, cloud and data consultancy, Kuala Lumpur	Discovery call with a prospect	Brief, fit map, proposal document, pricing
Enterprise	Karang Systems, systems integrator, Jakarta	Tender PDF or RFP email	Proposal draft; compliance matrix and bid decision next

The persona changes the intake parser and the output template. The research, fit and review agents are shared. Every demo seller is a synthetic tech company, one for each of the customer groups Google Cloud serves: startups, SMEs and enterprises.

3. Demo scenario (the three-minute video)

The video follows one opportunity end to end, then shows the same engine serving a second persona in a single cut. All prospect data comes from public sources and the seller's catalogue is synthetic, so the repository and the demo are safe to publish.

Time	Scene	What the judges see
0:00	The hub	Kelapa Cloud, a Kuala Lumpur cloud and data consultancy, opens its FlowHub Overview: what needs attention, what the agents did overnight. A calendar invite arrives: "Intro call, [retail chain], Thursday 10am". A consultant pastes it in.
0:20	Know yourself	A ten-second look at an offering in the hub: its rate card grid with source chips, proof, and the opportunities it has been used in. All extracted from the company's own deck and rate card.
0:35	Research brief	The Researcher agent streams a brief: what the retailer does, a recent expansion announcement, an e-commerce job posting, its current web stack. Each line carries a source chip; one is clicked.
1:05	Fit map	Gaps on the left, offerings on the right, lines between them. Two customisation ideas. One gap marked "cannot meet, consider partner". A go score of 78 with three reasons.
1:35	Proposal	The Writer opens a Google Doc: tailored executive summary, scope, indicative pricing from the rate card, case study. The Reviewer flags one unsupported claim, the consultant removes it, and the check reads "11 claims, 11 supported".
2:05	The team	A delivery engineer with Viewer access follows the deal room read-only, then asks the hub "What's our Growth price for 80 VMs?" and gets an answer cited to the rate card cell.
2:35	Same engine, new persona	Stoklane, a Singapore SaaS startup, pastes a restaurant group's pilot invite. The brief, the fit and a pilot proposal follow. A final frame shows the three sample sellers and the architecture.

> **Judging tip built into the demo.** The video shows the Reviewer rejecting an unsupported claim once, and the human editing the fit map before the proposal is written. Judges score meaningful AI incorporation and human-in-the-loop design higher than a fully automatic run.

4. Architecture

FlowHub is built entirely on Google Cloud. The web application runs on Cloud Run. The agents are written with the Agent Development Kit (ADK) and hosted on Agent Runtime in Gemini Enterprise Agent Platform (formerly Vertex AI Agent Engine). All state lives in Firestore, which also serves vector search for the offering graph. Documents are stored in Cloud Storage and read by Gemini directly.

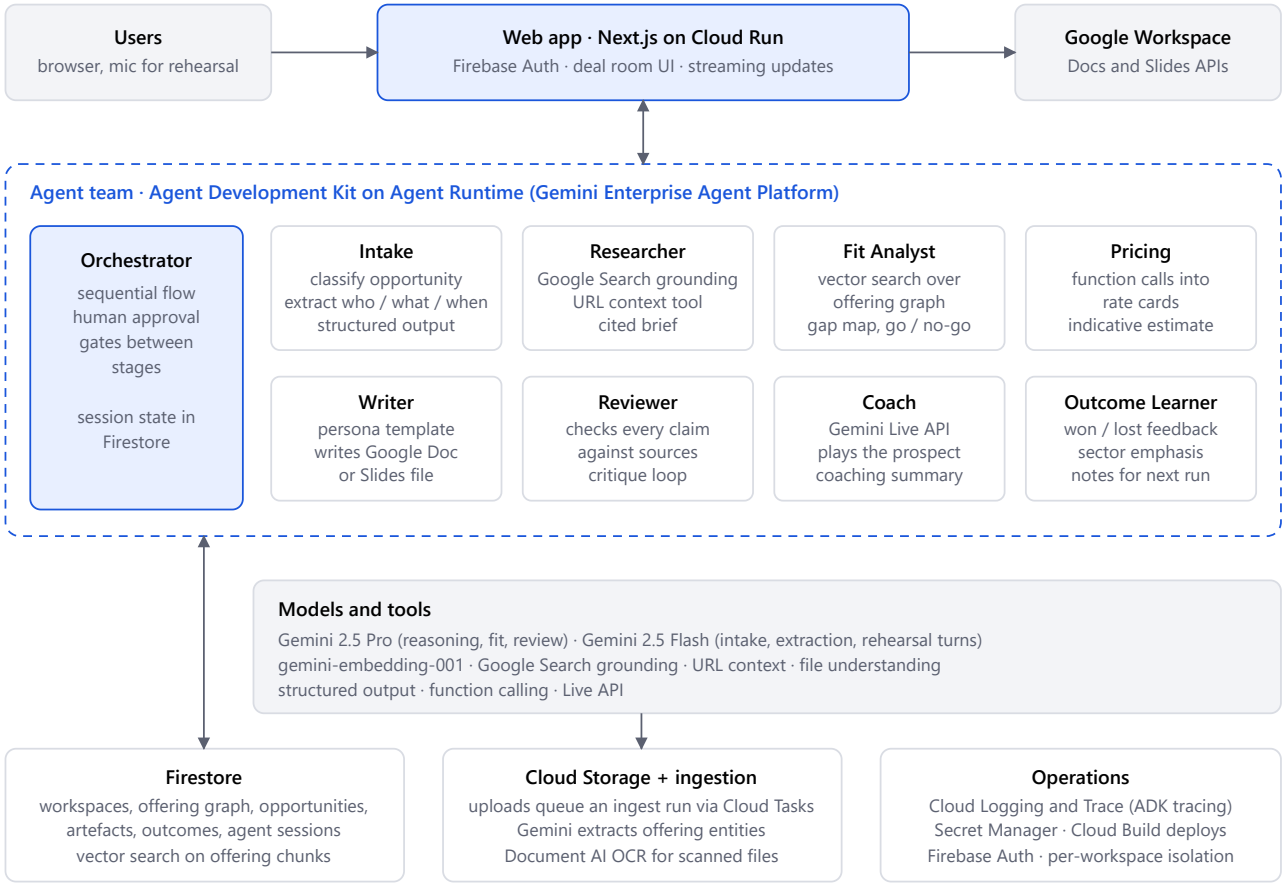


Figure 1. System architecture. Solid boxes are Google Cloud services; the dashed region is the ADK agent team hosted on Agent Runtime.

What makes it different

	Meeting-prep tools	Proposal writers	RFP suites	CRM AI	FlowHub
Researches the prospect	Yes	No	No	Yes	Yes, cited
Knows your priced offering	No	Templates	Content library	SKUs	Offering graph
Maps gaps to offerings	No	No	Partial	No	Yes
Produces the artefact	No	Yes	Yes	Partial	Docs and Slides
Rehearsal for this meeting	No	No	No	No	Yes
Works without a CRM	No	Yes	No	No	Yes

5. How it works in detail

Google Cloud and Google AI products used

Product	Role in FlowHub
Gemini 2.5 Pro and Flash (Gemini Enterprise Agent Platform)	Pro for research synthesis, fit reasoning and review; Flash for intake, extraction, streaming and rehearsal turns where latency matters.
Agent Development Kit (ADK)	Defines the six agents, their tools and the sequential orchestration with human approval gates. Gives built-in tracing and evaluation.
Agent Runtime (Gemini Enterprise Agent Platform)	Managed runtime for the ADK agents, with session state and scaling handled by the platform. Fallback: the same ADK app on a second Cloud Run service.
Google Search grounding, URL context	The Researcher reads the prospect's website and public signals live, and every sentence in the brief carries its grounding source.
Gemini file understanding, Document AI	Offering documents (PDF, PPTX, XLSX, images) go to Gemini directly; scanned PDFs pass through Document AI OCR first.
gemini-embedding-001, Firestore vector search	Offering graph chunks are embedded and retrieved by the Fit Analyst with a nearest-neighbour query scoped to the workspace.
Gemini Live API	Real-time voice for the rehearsal. The Coach agent receives the brief and fit map as system context and plays the prospect.
Cloud Run, Firebase Auth, Firestore, Cloud Storage, Cloud Tasks	Application hosting, sign-in, state and files. Cloud Tasks queues each agent stage and each ingestion run, so a run completes even if the browser closes.
Google Workspace APIs	The Writer creates the proposal as a Google Doc, or a deck outline as Slides, in the user's Drive.

The offering graph

Ingestion turns unstructured sales material into typed entities: *Offering* (product or service, with its features), *RateCard* (tiers or packages against line items), *FeeSchedule* (one-time, recurring and variable charges), *Proof* (case study, metric, logo), *Constraint* (what we do not do), and *Partner*. Past proposals are kept as source documents. Each entity, down to a single price cell, keeps a link to the page or slide it came from, so the Fit Analyst can cite the seller's own material the same way the Researcher cites the web. Gemini extracts entities with a JSON schema through structured output, and the user confirms or edits them in a review screen before they are used.

Request flow for one opportunity

1. The web app writes the opportunity to Firestore and starts an Agent Runtime session.
2. The Orchestrator runs Intake, then Researcher. Progress streams back to the deal room through Firestore listeners.
3. The human reads the brief and presses Continue. The Fit Analyst retrieves offerings by vector search and maps each gap. The Pricing agent then selects rate-card lines through function calls, and code copies the prices, so the model never writes a figure.
4. The human edits the fit map, then Writer produces the artefact and Reviewer critiques it. If Reviewer finds an unsupported claim, Writer revises, up to two loops.
5. The artefact link, the review report and the go/no-go score are saved on the opportunity. Rehearsal and outcome capture are available from here.

Controls against hallucination

- Research claims are only accepted with a grounding source; ungrounded sentences are dropped at the Researcher stage, not hidden.
- Prices and offerings come from function calls into the offering graph, never from the model's memory.
- The Reviewer scores each claim in the artefact as supported by a research source, an offering entity, or neither. "Neither" blocks the artefact until the human accepts or removes it.
- ADK evaluation sets with ten sample opportunities run in CI to catch regressions in fit quality.

Scalability

Every service is stateless and scales to zero: Cloud Run for the app, Agent Runtime for the agents, Firestore for state and vectors. Workspaces are isolated by Firestore security rules and by a workspace filter on every vector query. Adding a persona is a new intake parser and output template; adding a language is a prompt change, since Gemini handles JAPAC languages natively.

6. Risks

Risk	Mitigation
Research about a real company is wrong	Grounding required per claim, sources shown, Reviewer gate, and demo prospects chosen where public information is rich.
Agent Runtime setup costs time	Same ADK code runs on Cloud Run; decide by end of week 2.
Live API rehearsal is fragile in a demo	Stretch item; recorded once in the video, with a text fallback in the app.
Scope creep (hub, roles, three personas)	Roles stay light (three roles, no departments). Enterprise runs end to end with the standard proposal; its compliance matrix is a stretch item. Should items are cut first.
Google Cloud cost	Free tiers plus new-account credits cover the build; all services scale to zero.

Beyond the hackathon

FlowHub is priced per workspace, with a free tier for small teams. The hub becomes more valuable with every document, proposal and outcome recorded, which is the retention mechanism. The tender persona extends naturally into public procurement portals across the region.

7. Alignment with the judging criteria

Criterion	Evidence in FlowHub
Technical merit and Gen AI implementation (40%)	Six ADK agents with distinct tools; grounding, structured output, function calling, embeddings, file understanding and the Live API each used where they fit; a critique loop; evaluation in CI; fully serverless on Google Cloud.
Problem alignment and impact (25%)	Targets companies whose product knowledge is scattered and who sell without a large sales team. Gives the whole team one grounded source of truth, and turns a repeated, skipped preparation task into minutes.
Innovation and creativity (25%)	The company's own hub of what it sells, as the agents' grounded memory, gap-to-offering mapping, and code-enforced citations and pricing. No existing product joins these, and none works without a CRM.
User experience and solution design (10%)	One hub for the whole team with role-aware views, one deal room per opportunity, progress streamed live, approval gates instead of black-box automation, and outputs delivered as real Workspace files.

8. Build plan (22 September to 18 October)

Week	Milestone	Done when
1 · to 28 Sep	Project scaffold on Cloud Run with Firebase Auth. Offering ingestion: upload, Gemini extraction to schema, review screen, embeddings in Firestore.	A synthetic seller catalogue is loaded and searchable.
2 · to 5 Oct	ADK agent team on Agent Runtime: Intake, Researcher with grounding and citations, Fit Analyst with vector retrieval, Pricing functions. Deal room UI with streaming and approval gates.	Paste an invite, get a cited brief and a fit map.
3 · to 12 Oct	Writer into Google Docs, Reviewer critique loop, startup persona, hub members and roles. Evaluation set in CI. Rehearsal with the Live API if on schedule.	End-to-end run for three personas, reviewed artefact.
4 · to 18 Oct	Polish, outcome capture, video, deck, proposal PDF, public repository clean-up, final deployment check.	All deliverables submitted.

Must have: the hub with roles, ingestion, research with citations, fit map, one artefact into Docs, deployed on Cloud Run. **Stretch, in order:** ask the hub, rehearsal, Slides output, follow-up email, outcome learning, tender compliance matrix.

Team

Member	Focus
[Member 1]	Product, web application, deal room UI, ingestion, Workspace integration, video.
[Member 2]	ADK agent team, grounding and retrieval, Reviewer loop, evaluation, Agent Runtime deployment.